Final Projects

CS 485, Spring 2026
Applications of Natural Language Processing

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[Slides by Laure Thompson]

Final Projects

See the Final Projects page on Canvas under Pages

Project Overview

Investigate, analyze, and come to research findings about new methods, or insights on previously existing methods.

In groups of 2 - 3, you will either *build* a natural language processing system or *apply* one to some task.

Your project must: (1) use or develop a dataset, and

(2) report empirical results/analyses with this dataset

Project Components

Proposal: A 2 page document outlining the problem, your approach, possible dataset(s) and/or software systems to use.

Progress Report: A 4–8 page document that describes your preliminary work and results

Presentation: An opportunity to present your near-complete project to the class.

Final Report: An 8–12 page document that describes your project and final results.

Where to start

- What *core question(s)* are you trying to answer?
- How will you *operationalize* this question?
- What work are you building off of? What has been done before?
- What experiments will you run?
- How will you measure the success of these experiments?
  e.g., held - out accuracy, error analysis, manual evaluation, etc.

Where to look for related work?

NLP research papers:

- The ACL Anthology is a good place to start: https://aclanthology.org/
- Some Resources:
  - On how to read research papers, http://ccr.sigcomm.org/online/files/p83-keshavA.pdf
  - On navigating the NLP research space, https://www.junglelightspeed.com/the-top-10-nlp-conferences/

How to search for papers

- Search keywords in the ACL anthology, Google Scholar (https://scholar.google.com/), Semantic Scholar (https://www.semanticscholar.org/)
- Look at the papers that a paper references and those that cite it
- Examine other papers by a given author and their lab

Where to look for related work?

A standard web search can also be useful for finding…

- Research blog posts
- Datasets
- Related codebases
- Recorded Talks
- …and more!

Choice of emphasis

- Implementing and developing algorithms and features
- Defining a new linguistic / text analysis task, and tackling it with off - the- shelf NLP software
- Collect and explore a new textual dataset to address research hypotheses about it

A large variety of tasks

Detection Tasks

Classification Tasks

Prediction Tasks

- Predict external information from text (e.g. movie revenue, post popularity, stock volatility, etc.)

Structured Linguistic Prediction

- Relation, event extraction
- Narratives, character extraction

Visualization & Exploration

- Temporal analysis of events
- Topic modeling & clustering

Text Generation Tasks

- Machine Translation
- Summarization & Normalization
- Poetry / Lyric generation

Evaluation / Analysis

- Pre-existing LLM: strengths, biases, etc.

End-to-End Systems

- Question Answering
- Conversational dialogue systems

For more dataset and task ideas

- Shared task websites
  - SemEval: Series of semantic evaluation tasks.
    - SemEval tasks, 2025: https://aclanthology.org/events/semeval-2025/, 2024: https://aclanthology.org/events/semeval-2024/, etc.
      There may be access to data!
  - CoNLL shared tasks

- HuggingFace datasets website https://huggingface.co/docs/datasets/index
- Kaggle, https://www.kaggle.com/datasets

Some projects from recent years

Text Classification

- Song genre classification using lyrics
- Comparing models for multi - labeled classification of book genres
- Distinguishing between 19 th and 20 th century literature
- Predicting political slant in news comments
- Classification of political views on Reddit
- Classifying BBC news articles into their section/category types
- Language classification

Some projects from recent years

Detection Tasks

- Paraphrase detection
- Toxicity level detection in social media posts

Prediction Tasks

- Estimating stock volatility from news articles
- r/AmITheAsshole verdict prediction
- Predicting tweet popularity

Text Generation Tasks

- Text summarization for lectures

End-to-End Systems

- FAQ answering
- Medical diagnosis chatbot

Visualization & Exploration

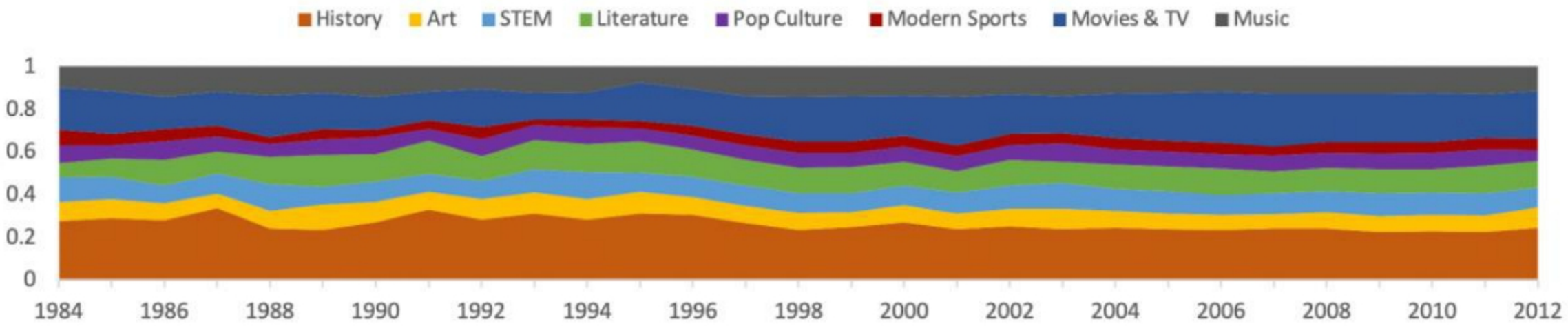
- Sentiment analysis of songs throughout time
- Sentiment analysis of r/wallstreetbets

JEOPARDY! Category Analysis

Evan Risas & Alisa Kotliarova

Task: Analyze each question-answer pair to determine which broad category it most closely fits, then predict category frequency for future Jeopardy games.

Dataset	Approach	Observe	Predict
200,000+ Jeopardy! Question & Answer pairs	Classify into custom categories using NLP model built on Word2Vec	Examine category popularity over time	Predict future category frequency



Brainstorming Session